Top Defect-Driving Parameters

Source of ranking: saved classifier importances from `models/best_classifier.pkl` and `models/feature_columns.pkl`. The saved best classifier is `GradientBoostingClassifier`, selected in `stage3_supervised_model.py`. PCA and Isolation Forest inputs come from `models/pca_feature_names.json` and `models/isolation_forest_feature_names.json`. Rule-based QA decisions come from `interpretation_rules.py` and `dashboard/risk_scoring.py`.

SHAP note: no SHAP implementation or SHAP artifact was found in the codebase. Explainability currently uses model `feature_importances_`, absolute coefficients where applicable, permutation importance in `dashboard/ml_evaluation.py`, or correlation fallback logic in `dashboard/pipeline.py` / `dashboard/charts.py`.

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
1	`sb_1`	Antimony chemistry 1	0.280921	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
2	`barinoc`	Barinoc inoculant/additive	0.200552	Defect classifier, PCA clustering, Isolation Forest	Numeric additive input. No explicit rule rationale found; importance comes from trained classifier.
3	`pouring_wt`	Pouring weight	0.042839	Defect classifier, PCA clustering, Isolation Forest	Numeric process input. No explicit rule rationale found; importance comes from trained classifier.
4	`silicon_ad`	Silicon	0.028261	Defect	Numeric

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
	`dition_1`	addition 1		classifier, PCA clustering, Isolation Forest	additive input. Silicon is also used by CE, C/Si, graphitization, shrinkage, and chemistry-stability feature logic.
5	`cr`	Chromium	0.028106	Defect classifier, PCA clustering, Isolation Forest	Used by graphitization feature logic as a carbide stabilizer/anti-graphitizer.
6	`ti`	Titanium	0.025903	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
7	`pouring_temp`	Pouring temperature	0.023035	Defect classifier, PCA clustering, Isolation Forest, `feat_temp_loss`, `feat_pouring_stability`, `feat_pouring_risk`, `feat_oxidation_risk`, `feat_shrinkage_risk_index`, QA temperature rules	Code flags cold pour, overheated pour, temperature loss, oxidation, shrinkage, cold shut, mis-run, gas porosity, and coarse grain risk.

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
8	`cu`	Copper	0.019741	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
9	`mn_1`	Manganese 1	0.017616	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. Bridged to Mn logic in upload aliases; Mn/S rules use manganese to neutralize sulfur.
10	`bath_s`	Bath sulfur	0.016747	Defect classifier, PCA clustering, Isolation Forest	Numeric sulfur-related input. Sulfur rules state high sulfur degrades nodularisation and consumes Mg treatment.
11	`p`	Phosphorus	0.014498	Defect classifier, PCA clustering, Isolation Forest	Used in CE calculation when bridged to `p__`; CE rules classify hypo/hypereutectic risk.
12	`s`	Sulfur	0.014159	Defect classifier, PCA clustering, Isolation Forest	Sulfur rules directly escalate risk and recommendations because sulfur compromises nodularisation and Mg treatment.

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
13	`ni`	Nickel	0.012766	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
14	`tapping_temp`	Tapping temperature	0.012588	Defect classifier, PCA clustering, Isolation Forest, `feat_temp_loss`, QA temperature rules	Used with pouring temperature to compute temperature loss; code links high loss to cold shuts, mis-runs, ladle heat loss, and transfer delays.
15	`sb`	Antimony	0.012120	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
16	`pouring_time_sec`	Pouring time	0.011934	Defect classifier, PCA clustering, Isolation Forest	Numeric process input. No explicit rule rationale found; importance comes from trained classifier.
17	`v`	Vanadium	0.011845	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
					found; importance comes from trained classifier.
18	`ce_1`	Cerium / CE variant 1	0.011718	Defect classifier, PCA clustering, Isolation Forest	Graphitization feature logic uses `ce_` as a graphite promoter when present.
19	`w`	Tungsten	0.011665	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
20	`al_1`	Aluminium 1	0.011381	Defect classifier, PCA clustering, Isolation Forest	Oxidation feature logic flags Al above 0.02 as oxide-inclusion/subsurface-pinhole risk when bridged to `al__`.
21	`nb`	Niobium	0.011009	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
22	`si`	Silicon	0.010752	Defect classifier, PCA clustering, Isolation	Silicon is used by CE, C/Si, graphitization, shrinkage, and

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
				Forest	chemistry-stability logic; low Si contributes to shrinkage index.
23	`c`	Carbon	0.010402	Defect classifier, PCA clustering, Isolation Forest	Carbon is used by CE, C/Si, and chemistry-stability logic; CE rules link off-target CE to shrinkage/hard spots or graphite flotation.
24	`sg_pigaddition_1`	SG pig addition 1	0.010399	Defect classifier, PCA clustering, Isolation Forest	Numeric charge/addition input. No explicit rule rationale found; importance comes from trained classifier.
25	`rr_1`	Return/revert ratio 1	0.010311	Defect classifier, PCA clustering, Isolation Forest	Numeric charge input. No explicit rule rationale found; importance comes from trained classifier.
26	`n`	Nitrogen	0.010036	Defect classifier, PCA clustering, Isolation Forest, `feat_gas_risk_index` when bridged to `n__`	Gas porosity feature logic flags elevated nitrogen as blowhole risk.

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
27	`zn`	Zinc	0.009163	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
28	`tapped_wt`	Tapped weight	0.008271	Defect classifier, PCA clustering, Isolation Forest	Numeric process input. No explicit rule rationale found; importance comes from trained classifier.
29	`fsm`	FSM additive	0.008021	Defect classifier, PCA clustering, Isolation Forest	Numeric nodulariser/additive input. FSM efficiency feature logic exists for `fsmaddition_mt`, but this raw `fsm` column is used directly by models.
30	`co`	Cobalt	0.007936	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
31	`rr`	Return/revert ratio	0.007611	Defect classifier, PCA clustering,	Numeric charge input. No explicit

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
				Isolation Forest	rule rationale found; importance comes from trained classifier.
32	`b`	Boron	0.007394	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
33	`sn`	Tin	0.006995	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
34	`lcrca_addition_1`	LCRCA addition 1	0.006161	Defect classifier, PCA clustering, Isolation Forest	Numeric charge/addition input. CRCA is used by gas-risk logic when column `crca` is present.
35	`zr`	Zirconium	0.005732	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
36	`sg_pig`	SG pig	0.005544	Defect	Numeric

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
				classifier, PCA clustering, Isolation Forest	charge input. Gas-risk comments identify SG Pig moisture as a gas-porosity driver, but formula uses `crca`, `n__`, and `mg_`.
37	`as`	Arsenic	0.005537	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
38	`ultraseed`	Ultraseed inoculant	0.005134	Defect classifier, PCA clustering, Isolation Forest	Numeric additive input. No explicit rule rationale found; importance comes from trained classifier.
39	`graphite`	Graphite addition	0.004732	Defect classifier, PCA clustering, Isolation Forest	Numeric carbon/addition input. CE rules recommend carbon/graphite adjustment when CE is low.
40	`low_re_fsm`	Low rare-earth FSM	0.004286	Defect classifier, PCA clustering, Isolation Forest	Numeric FSM/additive input. No explicit rule rationale found; importance

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
					comes from trained classifier.
41	`crca`	CRCA charge	0.004231	Defect classifier, PCA clustering, Isolation Forest, `feat_gas_risk_index`	Gas-risk logic treats CRCA above 50 as moisture/charge-related porosity risk.
42	`silicon_addition_2`	Silicon addition 2	0.003437	Defect classifier, PCA clustering, Isolation Forest	Numeric silicon additive input. Silicon participates in CE/graphitization/shrinkage logic through chemistry columns.
43	`pb`	Lead	0.003300	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
44	`fe`	Iron	0.003080	Defect classifier, PCA clustering, Isolation Forest	Numeric chemistry input. No explicit rule rationale found; importance comes from trained classifier.
45	`preseed`	Preseed inoculant	0.002754	Defect classifier, PCA clustering, Isolation Forest	Numeric additive input. No explicit rule rationale found;

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
					importance comes from trained classifier.
46	`ce`	Carbon equivalent	0.002581	Defect classifier, PCA clustering, Isolation Forest, `feat_ce_calculated`, CE rules	CE rules link low CE to micro-shrinkage/hard spots and high CE to graphite flotation/surface defects.
47	`sorel_addition`	Sorel addition	0.002175	Defect classifier, PCA clustering, Isolation Forest	Numeric charge/addition input. No explicit rule rationale found; importance comes from trained classifier.
48	`sorel_total`	Sorel total	0.001773	Defect classifier, PCA clustering, Isolation Forest	Numeric charge/addition input. No explicit rule rationale found; importance comes from trained classifier.
49	`last_heel_metal`	Last heel metal	0.000000	Defect classifier, PCA clustering, Isolation Forest	Numeric process/charge input. Heel ratio feature logic exists for `heel` and `tapped_wt_`; this raw column is used directly by models.
50	`sg_pigaddi`	SG pig	0.000000	Defect	Numeric

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
	`tion_2`	addition 2		classifier, PCA clustering, Isolation Forest	charge/addition input. No explicit rule rationale found; importance is zero in the saved classifier.
51	`heel`	Heel	0.000000	Defect classifier, PCA clustering, Isolation Forest	Heel ratio feature logic links high heel ratio to carry-over of S, N, and trace elements when `tapped_wt_` is available.
52	`cs`	CS addition	0.000000	Defect classifier, PCA clustering, Isolation Forest	Numeric additive input. No explicit rule rationale found; importance is zero in the saved classifier.
53	`al`	Aluminium	0.000000	Defect classifier, PCA clustering, Isolation Forest	Oxidation feature logic flags Al above 0.02 when bridged to `al__`; this raw column has zero saved classifier importance.
54	`high_re_fsm`	High rare-earth FSM	0.000000	Defect classifier, PCA clustering, Isolation Forest	Numeric FSM/additive input. No explicit rule rationale found; importance is zero in the

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
					saved classifier.
55	`mn`	Manganese	0.000000	Defect classifier, PCA clustering, Isolation Forest	Manganese is used in Mn/S and graphitization logic when available as `mn_`; this raw column has zero saved classifier importance.
56	`zero_re_fsm`	Zero rare-earth FSM	0.000000	Defect classifier, PCA clustering, Isolation Forest	Numeric FSM/additive input. No explicit rule rationale found; importance is zero in the saved classifier.
57	`superseed`	Superseed inoculant	0.000000	Defect classifier, PCA clustering, Isolation Forest	Numeric additive input. No explicit rule rationale found; importance is zero in the saved classifier.
58	`reseed`	Reseed inoculant	0.000000	Defect classifier, PCA clustering, Isolation Forest	Numeric additive input. No explicit rule rationale found; importance is zero in the saved classifier.
59	`fsmaddition_mt`	FSM addition per metric ton	N/A	PCA clustering, Isolation	FSM efficiency logic

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
				Forest, `feat_fsm_s_index`, `feat_fsm_undertreat_risk` when available	normalizes FSM dose by sulfur; low FSM with high sulfur is under-treatment risk. Excluded from supervised classifier by `stage3_supervised_model.py`.
signal	`defect_prob`	Defect probability	N/A	Produced by classifier in `dashboard/pipeline.py`; consumed by `dashboard/risk_scoring.py` and `interpretation_rules.py`	Thresholds drive MONITOR, HOLD, and STOP decisions.
signal	`anomaly_score`	Anomaly score	N/A	Produced by Isolation Forest/LOF in `stage5_anomaly_detection.py` or dashboard inference; consumed by risk scoring and QA rules	Thresholds drive MEDIUM, HIGH, and CRITICAL risk escalation for unusual process patterns.
signal	`cluster`	Cluster label	N/A	Produced by KMeans in `stage4_pca_clustering.py`; consumed by `compute_cluster_stats()` in `dashboard/`	Cluster historical defect rate can escalate HOLD or STOP decisions.

Rank	Column name	Human-readable name	Classifier importance	Where used in pipeline	Why it influences defects according to code
				risk_scoring.py	

Top Defect-Driving ML Features

The table below is the complete feature union actually used by saved ML artifacts. `Classifier` means `models/best_classifier.pkl`; `PCA` means `models/pca_model.pkl`; `Isolation Forest` means `models/isolation_forest.pkl`. Classifier importance is unavailable for `fsmaddition_mt` because the supervised training code excludes it.

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
`sb_1`	0.280921	Y	Y	Y	Raw numeric input selected in `stage3_supervised_model.py`	Antimony chemistry 1	Learned by classifier; highest saved importance. Direction not encoded in tree importance.
`barinoc`	0.200552	Y	Y	Y	Raw numeric input	Barinoc inoculant/additive	Learned by classifier; second-highest saved importance. Direction not encoded in tree importance.
`pouring_wt`	0.042839	Y	Y	Y	Raw numeric input	Pouring weight	Learned by classifier; direction not encoded in tree

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							importance.
silicon_addition_1	0.028261	Y	Y	Y	Raw numeric input	Silicon addition 1	Learned by classifier; silicon also affects engineered CE/chemistry features when present as chemistry.
cr	0.028106	Y	Y	Y	Raw numeric input	Chromium	Learned by classifier; graphitization logic treats chromium as anti-graphitizer when bridged to cr_.
ti	0.025903	Y	Y	Y	Raw numeric input	Titanium	Learned by classifier; direction not encoded in tree importance.
pouring_temp	0.023035	Y	Y	Y	Raw input; also used in feat_temp_loss, feat_pouring_stability, feat_pouring_risk, feat_oxidation	Pouring temperature	Off-range pouring temperature and high thermal loss increase rule-based risk; classifier also learned this raw input.

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
					_risk, feat_shrinkage_risk_index		
cu	0.019741	Y	Y	Y	Raw numeric input	Copper	Learned by classifier; direction not encoded in tree importance.
mn_1	0.017616	Y	Y	Y	Raw numeric input	Manganese 1	Learned by classifier; manganese is related to Mn/S sulfur neutralization logic when bridged.
bath_s	0.016747	Y	Y	Y	Raw numeric input	Bath sulfur	Learned by classifier; sulfur-related rules link sulfur to nodularisation/Mg treatment risk.
p	0.014498	Y	Y	Y	Raw numeric input	Phosphorus	Learned by classifier; CE formula uses phosphorus when bridged to p___.
s	0.014159	Y	Y	Y	Raw numeric input	Sulfur	Learned by classifier; sulfur rules

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							directly escalate risk and recommendations.
`ni`	0.012766	Y	Y	Y	Raw numeric input	Nickel	Learned by classifier; direction not encoded in tree importance.
`tapping_temp`	0.012588	Y	Y	Y	Raw input; used in `feat_temp_loss`	Tapping temperature	Drives temperature-loss feature and QA temperature rules.
`sb`	0.012120	Y	Y	Y	Raw numeric input	Antimony	Learned by classifier; direction not encoded in tree importance.
`pouring_time_sec`	0.011934	Y	Y	Y	Raw numeric input	Pouring time	Learned by classifier; direction not encoded in tree importance.
`v`	0.011845	Y	Y	Y	Raw numeric input	Vanadium	Learned by classifier; direction not encoded in tree importance.
`ce_1`	0.011718	Y	Y	Y	Raw	Cerium /	Graphitizat

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
					numeric input	CE variant 1	ion logic uses `ce_` as graphite promoter when present.
`w`	0.011665	Y	Y	Y	Raw numeric input	Tungsten	Learned by classifier; direction not encoded in tree importance.
`al_1`	0.011381	Y	Y	Y	Raw numeric input	Aluminium 1	Oxidation logic uses aluminium when bridged to `al__`.
`nb`	0.011009	Y	Y	Y	Raw numeric input	Niobium	Learned by classifier; direction not encoded in tree importance.
`si`	0.010752	Y	Y	Y	Raw numeric input	Silicon	Affects CE, graphitization, shrinkage, and chemistry-stability features when bridged.
`c`	0.010402	Y	Y	Y	Raw numeric input	Carbon	Affects CE, C/Si ratio, and chemistry-stability features

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							when bridged.
`sg_pigaddition_1`	0.010399	Y	Y	Y	Raw numeric input	SG pig addition 1	Learned by classifier; direction not encoded in tree importance.
`rr_1`	0.010311	Y	Y	Y	Raw numeric input	Return/ revert ratio 1	Learned by classifier; direction not encoded in tree importance.
`n`	0.010036	Y	Y	Y	Raw numeric input	Nitrogen	Gas-risk logic uses nitrogen when bridged to `n__`; high N contributes to blowhole risk.
`feat_temp_loss`	0.009802	Y	Y	Y	`tapping_temp - pouring_temp`; `stage2_feature_engineering.py`	Temperature loss	Higher loss flags ladle heat loss/cold-pour risk in rules.
`zn`	0.009163	Y	Y	Y	Raw numeric input	Zinc	Learned by classifier; direction not encoded in tree importance

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							.
`tapped_wt`	0.008271	Y	Y	Y	Raw numeric input	Tapped weight	Learned by classifier; direction not encoded in tree importance.
`fsm`	0.008021	Y	Y	Y	Raw numeric input	FSM additive	Learned by classifier; FSM dose logic exists for `fsmaddition_mt`.
`co`	0.007936	Y	Y	Y	Raw numeric input	Cobalt	Learned by classifier; direction not encoded in tree importance.
`rr`	0.007611	Y	Y	Y	Raw numeric input	Return/ revert ratio	Learned by classifier; direction not encoded in tree importance.
`b`	0.007394	Y	Y	Y	Raw numeric input	Boron	Learned by classifier; direction not encoded in tree importance.
`sn`	0.006995	Y	Y	Y	Raw numeric input	Tin	Learned by classifier; direction not encoded in

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							tree importance.
lcrca_addition_1	0.006161	Y	Y	Y	Raw numeric input	LCRCA addition 1	Learned by classifier; CRCA charge is also used by gas-risk logic via crca.
zr	0.005732	Y	Y	Y	Raw numeric input	Zirconium	Learned by classifier; direction not encoded in tree importance.
sg_pig	0.005544	Y	Y	Y	Raw numeric input	SG pig	Learned by classifier; gas-risk comments mention SG Pig moisture as porosity driver.
as	0.005537	Y	Y	Y	Raw numeric input	Arsenic	Learned by classifier; direction not encoded in tree importance.
ultraseed	0.005134	Y	Y	Y	Raw numeric input	Ultraseed inoculant	Learned by classifier; direction not encoded in tree importance.
graphit	0.004732	Y	Y	Y	Raw	Graphite	Carbon/CE

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
`e`					numeric input	addition	adjustment is referenced by CE rules; raw direction learned by classifier.
`low_re_fsm`	0.004286	Y	Y	Y	Raw numeric input	Low rare-earth FSM	Learned by classifier; direction not encoded in tree importance.
`crca`	0.004231	Y	Y	Y	Raw input; used in `feat_gas_risk_index`	CRCA charge	CRCA above 50 contributes to gas porosity risk in feature logic.
`silicon_addition_2`	0.003437	Y	Y	Y	Raw numeric input	Silicon addition 2	Learned by classifier; silicon chemistry drives several engineered features when present.
`pb`	0.003300	Y	Y	Y	Raw numeric input	Lead	Learned by classifier; direction not encoded in tree importance.
`fe`	0.003080	Y	Y	Y	Raw numeric input	Iron	Learned by classifier; direction

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							not encoded in tree importance.
`preseed`	0.002754	Y	Y	Y	Raw numeric input	Preseed inoculant	Learned by classifier; direction not encoded in tree importance.
`ce`	0.002581	Y	Y	Y	Raw CE input; copied to `feat_ce_calculated` if C/Si/P unavailable	Carbon equivalent	CE rules classify hypo/hyper eutectic risk.
`feat_ce_calculated`	0.002327	Y	Y	Y	`c_ + (si_ + p__) / 3`, else `ce`; `stage2_feature_engineering.py`	Carbon equivalent feature	Off-target CE drives shrinkage/ hard spots or graphite flotation/surface defects in rules.
`sorel_addition`	0.002175	Y	Y	Y	Raw numeric input	Sorel addition	Learned by classifier; direction not encoded in tree importance.
`sorel_total`	0.001773	Y	Y	Y	Raw numeric input	Sorel total	Learned by classifier; direction not encoded in tree

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							importance.
feat_ce_optimal	0.000416	Y	Y	Y	1 when CE is 4.2 to 4.3, else 0	Eutectic CE zone flag	Optimal CE zone is expected by code comments to be favorable.
feat_temp_loss_risk	0.000146	Y	Y	Y	1 when feat_temp_loss > 80	High temperature-loss flag	High loss is linked to cold shuts, mis-runs, and ladle heat loss.
feat_ce_hypo_risk	0.000079	Y	Y	Y	1 when CE < 4.2	Hypoeutectic CE risk	Code links low CE to micro-shrinkage and hard spots.
feat_gas_risk_index	0.000050	Y	Y	Y	2*(n__ > 0.008) + 1*(mg_ > 0.055) + 0.5*(crca > 50)	Gas porosity index	High N, high Mg, and CRCA moisture/charge risk contribute to blowholes or Mg vapor pockets.
feat_shrinkage_risk_index	0.000021	Y	Y	Y	1.5*(CE < 4.2) + 1*(si_ < 2.0) + 0.5*(pouring_temp > 1420)	Shrinkage risk index	Code links hypoeutectic CE, low Si, and high pouring temp to reduced graphite expansion/shrinkage risk.
feat_ce	0.000005	Y	Y	Y	1 when CE	Hypereute	Code links

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
`_hyper_risk`					> 4.3	ctic CE risk	high CE to graphite flotation and surface defects.
`last_heel_metal`	0.000000	Y	Y	Y	Raw numeric input	Last heel metal	Learned importance is zero; no direct rule rationale found for this exact column.
`sg_pigaddition_2`	0.000000	Y	Y	Y	Raw numeric input	SG pig addition 2	Learned importance is zero; no explicit rule rationale found.
`heel`	0.000000	Y	Y	Y	Raw numeric input	Heel	Heel-ratio logic says high heel can carry over S, N, and trace elements when ratio inputs are available.
`cs`	0.000000	Y	Y	Y	Raw numeric input	CS addition	Learned importance is zero; no explicit rule rationale found.
`al`	0.000000	Y	Y	Y	Raw numeric input	Aluminium	Oxidation logic uses aluminium when bridged to `al__`; this raw feature

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							has zero classifier importance.
`high_re_fsm`	0.000000	Y	Y	Y	Raw numeric input	High rare-earth FSM	Learned importance is zero; no explicit rule rationale found.
`mn`	0.000000	Y	Y	Y	Raw numeric input	Manganese	Mn/S and graphitization rules use manganese when available as `mn_`; this raw feature has zero classifier importance.
`zero_re_fsm`	0.000000	Y	Y	Y	Raw numeric input	Zero rare-earth FSM	Learned importance is zero; no explicit rule rationale found.
`superseed`	0.000000	Y	Y	Y	Raw numeric input	Superseed inoculant	Learned importance is zero; no explicit rule rationale found.
`reseed`	0.000000	Y	Y	Y	Raw numeric input	Reseed inoculant	Learned importance is zero; no explicit rule rationale

Feature name	Classifier importance	Classifier	PCA	Isolation Forest	Formula or calculation source	Business meaning	Expected effect on defect prediction
							found.
`feat_pouring_stability`	0.000000	Y	Y	Y	`-1` if pouring temp < 1300; `1` if > 1500; else `0`	Pouring temperature stability	Off-range pouring temperature is coded as cold-pour or oxidation risk.
`feat_temp_loss_low`	0.000000	Y	Y	Y	`1` when `feat_temp_loss < 20`	Low temperature-loss flag	Code comments describe low loss as possible inaccurate measurement or very short ladle time.
`feat_pouring_risk`	0.000000	Y	Y	Y	`1` when `feat_pouring_stability != 0`	Pouring temperature out-of-range flag	Flags cold or overheated pouring conditions.
`feat_oxidation_risk`	0.000000	Y	Y	Y	`(al__ > 0.02) + 0.5*(pouring_temp > 1450)`	Oxidation risk index	Code links high Al and high pouring temp to oxide inclusions/ subsurface pinholes.
`feat_white_iron_risk`	0.000000	Y	Y	Y	`1` when `feat_graphitization_index < 0`	White iron risk flag	Negative graphitization tendency indicates carbide/white-iron risk.
`feat_graphitization_index`	0.000000	Y	Y	Y	`2*si_ + 5*ce_ - 3.5*cr__`	Graphitization tendency	Higher is graphitic tendency; lower is

Feature name	**Classifier importance**	**Classifier**	**PCA**	**Isolation Forest**	**Formula or calculation source**	**Business meaning**	**Expected effect on defect prediction**
					`0.3*mn_ - 3*mo_`		carbide/white-iron tendency per code comments.
`feat_chemistry_instability`	0.000000	Y	Y	Y	Sum of deviations outside target ranges for C, Si, Mn, Mg	Chemistry instability	Larger value means farther from ideal chemistry range.
`feat_chemistry_ok`	0.000000	Y	Y	Y	`1` when `feat_chemistry_instability < 0.1`	Chemistry OK flag	Indicates chemistry is close to coded target ranges.
`fsmaddition_mt`	N/A		Y	Y	Raw numeric input; also used by optional `feat_fsm_s_index` and `feat_fsm_undertreat_risk` formulas	FSM addition per metric ton	Used by unsupervised PCA/anomaly models; supervised classifier explicitly excludes it.

Parameters Most Responsible For HOLD Decisions

These are rule-traced HOLD drivers, not inferred correlations.

Driver	**Source columns/features**	**Code threshold/path**	**Resulting decision behavior**
High defect probability	`defect_prob`	`interpretation_rules.py`: `defect_prob >= 0.50`; `dashboard/risk_scoring.py`:	Escalates to HIGH risk and HOLD.

Driver	Source columns/features	Code threshold/path	Resulting decision behavior
		`defect_prob >= 0.50`	
Elevated anomaly score	`anomaly_score`	`interpretation_rules.py`: `anomaly_score >= 0.55`; `dashboard/risk_scoring.py`: `anomaly_score >= 0.60`	Escalates to HIGH risk and HOLD.
Hypoeutectic carbon equivalent	`feat_ce_calculated` or `ce`	`CE < 4.20`	Escalates to HIGH risk and HOLD; recommendation is to increase carbon/graphite/carburiser additions.
Cold pouring temperature	`pouring_temp`	`pouring_temp < 1290`	Escalates to HIGH risk and HOLD due to incomplete filling/cold shuts.
Elevated sulfur	`s_`	`s_ > 0.015` and `<= 0.025`	Escalates to HIGH risk and HOLD; recommendation is FSM/re-treatment monitoring.
Risky cluster history	`cluster`, `defect`, `defect_prob`	Cluster defect rate >= 0.22 in `dashboard/risk_scoring.py`	Escalates to HIGH risk and HOLD because the process group has elevated historical risk.

Parameters Most Responsible For STOP Decisions

Driver	Source columns/features	Code threshold/path	Resulting decision behavior
Critical defect probability	`defect_prob`	`defect_prob >= 0.75`	STOP and CRITICAL risk.
Critical anomaly score	`anomaly_score`	`dashboard/risk_scoring.py`: `anomaly_score >= 0.80`; `interpretation_rules.py`: `anomaly_score >= 0.80`	STOP and CRITICAL risk for radically different process pattern.

Driver	Source columns/features	Code threshold/path	Resulting decision behavior
Critical sulfur	`s_`	`s_ > 0.025`	STOP and CRITICAL risk due to severe nodularisation failure.
Severe low Mg recovery	`mg_recovery_`	`mg_recovery_ < 0.35` in rules	STOP and CRITICAL risk; code cites flake/vermicular graphite risk.
Critical cluster history	`cluster`, `defect`, `defect_prob`	Cluster defect rate >= 0.40 in `dashboard/risk_scoring.py`	STOP/CRITICAL escalation due to historical defect rate in process cluster.

Parameters Most Responsible For Critical Risk Classification

Driver	Source columns/features	Critical threshold/path	Notes
Defect probability	`defect_prob` from classifier	>= 0.75	Direct CRITICAL risk in both QA rules and unified risk scoring.
Anomaly score	`anomaly_score` from Isolation Forest/LOF or dashboard inference	>= 0.80	Direct CRITICAL risk in unified scoring and rules.
Sulfur	`s_`	> 0.025	Direct CRITICAL risk because sulfur depletes Mg treatment and degrades nodularisation.
Mg recovery	`mg_recovery_`	< 0.35	Direct CRITICAL risk in QA rules. Note: current saved classifier feature schema contains `mg_recovery`, not `mg_recovery_`; upload bridging can create `mg_recovery_` for rules/features.
Cluster historical defect rate	`cluster` plus `defect` or `defect_prob`	>= 0.40	Direct CRITICAL risk in unified scoring.

Executive Summary For Metallurgy Team

The strongest saved classifier drivers are `sb_1`, `barinoc`, `pouring_wt`, `silicon_addition_1`, `cr`, `ti`, and `pouring_temp`. These rankings come from the persisted Gradient Boosting model, not from manual interpretation.

The strongest rule-based HOLD/STOP drivers are different: `defect_prob`, `anomaly_score`, sulfur (`s_`), Mg recovery (`mg_recovery_`), carbon equivalent (`ce` / `feat_ce_calculated`), pouring temperature, and cluster historical defect rate. These directly change `risk_level` and `recommendation` through `interpretation_rules.py` and `dashboard/risk_scoring.py`.

The ML system uses three feature sets:

- Supervised defect classifier: 74 numeric features from `models/feature_names.json`.
- PCA/KMeans clustering: the same features plus `fsmaddition_mt`.
- Isolation Forest anomaly model: the same 75-feature set as PCA/KMeans.

The engineered metallurgical features actually present in the saved feature schemas are CE-zone flags, temperature-loss features, pouring-temperature risk flags, oxidation risk, graphitization/white-iron risk, shrinkage risk, gas risk, and chemistry-instability flags. Some stage-2 functions can create additional features (`feat_mg_recovery_risk`, `feat_mn_s_ratio`, `feat_sulfur_risk`, `feat_fsm_s_index`, etc.) when their canonical source columns exist, but those features are not in the current saved classifier/PCA/Isolation Forest feature schemas and therefore are not listed as current model inputs.